# *HRPO Observing Proposal*

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*Title:* Obtaining the color-magnitude diagram of star clusters by estimating the distance, age, and metallicity of the cluster.

## *Target List*

- **NGC 7089**
  RA: 21h 33m 27.02s
  DEC: –00° 49′ 23.7″
  Apparent Magnitude: 6.5
- **NGC 7492**
  RA: 23h 08m 26.7s
  DEC: −15° 36′ 39″
  Apparent Magnitude: 11.2
- **NGC 7099**
  RA: 21h 40m 22.12s
  Dec: –23° 10′ 47.5″
  Apparent Magnitude: 7.2

## *Estimate of total required telescope time*

Total observing time: 7 hours 1 minute

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## *Scientific Justification* -

Color magnitude diagram (CMD) is a plot of observational data of a population of stars. CMD shows us the plot of brightness (or luminosity) against color (surface temperature) of the stars. The plot of luminosity against temperature is also known as Hertzsprung-Russell (or H-R) diagram [1][2]. These theoretical diagrams have been reproduced for stellar populations such as open and globular clusters and even for galaxies. It is one essential tool for classifying stellar types including their properties. We can estimate a star's internal structure and evolutionary stage simply by determining its position on the diagram. Most stars, including the Sun, are in a band which runs from the top-left to the bottom-right of the chart [Figure: 1]. This band contains stars which are in their main sequence stage. We can use the chart to see the temperature of main sequence stars increases with brightness. Giant and supergiant stars are on the top and top-right of the diagram. This tells us they are brighter than main sequence stars but also redder and cooler. This is because they expand and cool as they reach the final stages of their lives. However, because of their large size, they remain very bright. White dwarf stars are below and to the left of the main sequence. This tells us they are hotter than main sequence stars, but not as bright. This is because they are smaller but contain a lot of mass. The diagram can be used by scientists to roughly measure how far away a star cluster or galaxy is from Earth. This can be done by comparing the apparent magnitudes of the stars in the cluster to the absolute magnitudes of stars with known distances (or of model stars). This technique is known as main sequence fitting and is a type of spectroscopic parallax [3]. As we can infer from the discussion above the color magnitude diagram is an essential tool for understanding the stellar evolution and various different properties like star formation histories, luminosity functions, chemical abundance, or metallicity etc. of the stars.

In our study we are attempting to create the color-magnitude diagram globular clusters. Globular clusters are very massive objects that contain hundreds of thousands or perhaps a million stars. If we observe a typical color magnitude diagram [Figure 2] we can see there are no Main Sequence stars of types OBAF, but there are many red giants. The brightest stars in a globular cluster are those at the tip of the red giant branch, which explains the red appearance of the bright stars in color images of the clusters. We can also see stars populating the horizontal branch, the asymptotic giant branch, and even some stars that have colors and magnitudes of F stars, but far fewer than the G stars just below and to the right of them on the Main Sequence.

Our project involves observation of 3 globular clusters NGC 7089, NGC 7492 and NGC 7099 and plotting the color magnitude diagram for each. We have chosen these globular clusters because they are the most feasible in terms of observation. As discussed above color magnitude diagram are very useful tools. Being limited to a ground-based telescope and limited observing time, we intent to create at least a partial color magnitude diagram as we will not be able to more than 20 in stellar magnitude due to restricted observing time. By creating the color magnitude diagram, we can, as a proof of concept show that we can in fact create a CMD using a ground-based telescope with limited observing time. Our color magnitude diagram can be used to study the evolution of the stars in globular clusters and also various stellar properties.

## *Experimental Design -*

As we can infer from the name, the creation of CMD requires the Color of the stars. From the numerous pictures taken from our observation and several code packages we can get the color of each star. We need to use the Data Reduction, Plate solving and the Photometry packages to resolve the images. And also, Image stacking packages to stack the numerous images taken over the observing period.
Our images will contain a lot of stars. As a result, finding the color of each star separately will be time consuming task. For this reason we intent to look for an alternative method, which can give us the required data for all the star from the images.

Another requirement for the CMD is the Absolute magnitude. Using archival data and also spectroscopic parallax we can find the flux of each star and the distance to the cluster, which can be used to calculate the absolute magnitude of each star. Using Color and the absolute magnitude the CMD can be created.

The total observing time we requested is 7 hours 1 minute. We used the LCO Exposure Time Calculator [4]. To use LCO we need the signal to noise ratio. After some literature review [5] we found we need signal to noise ratio of 10 to observe globular clusters. With S/N of 10 we need to estimate the magnitude we aim for. Using a suitable magnitude so that our observing time remains within limit, we are planning on following observing time:

- **NGC 7089:** Total observing time required:
  R band: 748s B band: 1539s and V band: 1120s with magnitude 19.2
  For each band we plan to observe each night with exposure of 60s with 3 repeats totaling 180s.
  As R band requires less time, we only need to observe in R band for 5 nights. And for rest of the nights, we plan to dedicate the R band time to the B band.

- **NGC 7099:** Total observing time required:
  R band: 894s B band: 1834s and V band: 1338s with magnitude 19.3
  For each band we plan to observe each night with exposure of 60s with 3 repeats totaling 180s.
  As R band requires less time, we only need to observe in R band for 5 nights. And for rest of the nights, we plan to dedicate the R band time to the B and V band.

- **NGC 7492:** Total observing time required:
  R band: 3900s and V band: 3900s with magnitude 20.5
  For each band we plan to observe each night with exposure of 60s with 5 repeats totaling 300s.
  As the target is fainter, we dedicate half the total observing time to it and only observe in V and R band over 8 nights.

## *References & Figures*


[1] Hertzprung, Ejnar (1908). "Über die Sterne der Unterabteilung c und ac nach der Spektralklassifikation von Antonia C. Maury". Astronomische Nachrichten. 179 (24): 373–380. Bibcode:1909AN....179..373H. doi:10.1002/asna.19081792402.

[2] Russell, Henry Norris (1914). "Relations Between the Spectra and Other Characteristics of the Stars". Popular Astronomy. 22: 275–294. Bibcode:1914PA.....22..275R.

[3] Carroll, Bradley W.; Ostlie, Dale A. (2017). An introduction to modern astrophysics (Second ed.). Cambridge, United Kingdom. p. 475. ISBN 9781108422161.

[4] https://exposure-time-calculator.lco.global/

[5] Sarajedini, Ata, et al. "The ACS survey of galactic globular clusters. I. Overview and clusters without previous Hubble Space Telescope photometry." The Astronomical Journal 133.4 (2007): 1658.

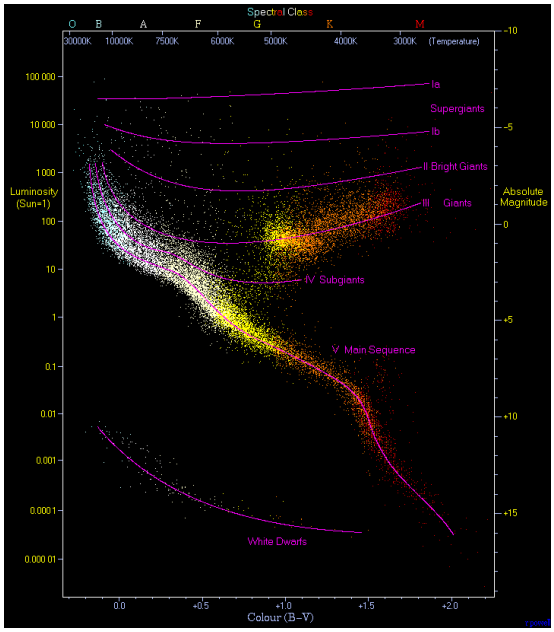




**Figure 1**: An observational H–R diagram with 22,000 stars plotted from the Hipparcos Catalogue and 1,000 from the Gliese Catalogue of nearby stars.

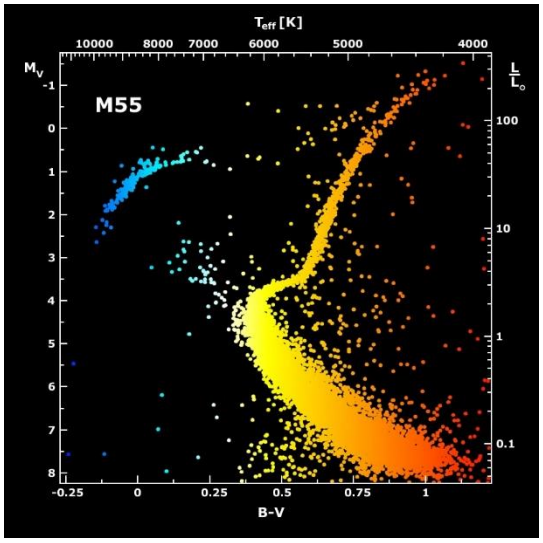



**Figure 2**: Color magnitude diagram of M55. Source: https://apod.nasa.gov/apod/ap010223.html